

How `predict-all-59.R` Extrapolates to Unknown Locations

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1 Purpose

This note translates the current implementation in the prediction script `predict-all-59.R` and the helper `predictY_cont_lambda()` from `update_params.R` into matrix notation. The goal is to explain, in mathematical terms, how a model that has already been fitted on observed AQS sites is pushed forward to unobserved CMAQ grid cells.

For setting 59, the fitted model is loaded from `results/us-all-full-59.RData`. In the current checkout, the prediction script uses:

$$\begin{aligned} n_s &= 1089 \quad \text{observed sites,} & n_p &= 2632 \quad \text{prediction cells,} & n_t &= 31 \quad \text{days,} \\ n_{\text{reps}} &= 5000 \quad \text{posterior draws,} & K &= 6 \quad \text{spatial partitions (knots).} \end{aligned}$$

Setting 59 is a symmetric t model with `skew = FALSE`, so the skew term is inactive:

$$\lambda = 0, \quad z_{k,t} = 0 \quad \text{for all } k, t.$$

Therefore the prediction step is driven by the regression mean, the spatial correlation matrix, and the partition-specific precision parameters.

2 Observed and Unknown Locations

Let

$$S_o = \begin{bmatrix} (s_1^o)^\top \\ \vdots \\ (s_{n_s}^o)^\top \end{bmatrix} \in \mathbb{R}^{n_s \times 2}$$

be the observed AQS locations after removing sites with more than 50% missing responses, and let

$$S_p = \begin{bmatrix} (s_1^p)^\top \\ \vdots \\ (s_{n_p}^p)^\top \end{bmatrix} \in \mathbb{R}^{n_p \times 2}$$

be the unobserved grid cells in the southeast subdomain selected by the code.

The script computes three distance matrices:

$$\begin{aligned} D_{22} &\in \mathbb{R}^{n_s \times n_s}, & (D_{22})_{ab} &= \|s_a^o - s_b^o\|_2, \\ D_{11} &\in \mathbb{R}^{n_p \times n_p}, & (D_{11})_{ab} &= \|s_a^p - s_b^p\|_2, \\ D_{12} &\in \mathbb{R}^{n_p \times n_s}, & (D_{12})_{ab} &= \|s_a^p - s_b^o\|_2. \end{aligned}$$

These matrices are the only spatial inputs used by the prediction step.

3 CMAQ Covariates and Design Matrices

The raw CMAQ field is centered and scaled once:

$$\tilde{C}(s, t) = \frac{C(s, t) - \bar{C}}{s_C},$$

where $\bar{C}$ and s_C are the global mean and standard deviation used by the script.

For each day $t \in \{1, \dots, n_t\}$, define the observed-site and prediction-site design matrices

$$X_{o,t} = \begin{bmatrix} 1 & \tilde{C}(s_1^o, t) \\ \vdots & \vdots \\ 1 & \tilde{C}(s_{n_s}^o, t) \end{bmatrix} \in \mathbb{R}^{n_s \times 2},$$

$$X_{p,t} = \begin{bmatrix} 1 & \tilde{C}(s_1^p, t) \\ \vdots & \vdots \\ 1 & \tilde{C}(s_{n_p}^p, t) \end{bmatrix} \in \mathbb{R}^{n_p \times 2}.$$

The regression coefficient vector from posterior draw i is

$$\beta_i = \begin{bmatrix} \beta_{0,i} \\ \beta_{1,i} \end{bmatrix} \in \mathbb{R}^2.$$

Hence the code-level regression means are

$$\mu_{i,t}^o = X_{o,t} \beta_i \in \mathbb{R}^{n_s}, \quad \mu_{i,t}^p = X_{p,t} \beta_i \in \mathbb{R}^{n_p}.$$

4 `simple.cov.sp()` in Mathematical Form

The function `simple.cov.sp()` receives a distance matrix $D = (d_{ab})$ and returns a covariance matrix $V = (V_{ab})$. Write

$$\text{sp.par} = (\sigma^2, \rho),$$

where σ^2 is a variance scale and ρ is a range parameter.

The returned covariance is

$$V_{ab} = C_0(d_{ab}) + \mathbf{1}(d_{ab} = 0)(\tau_{\text{fine}}^2 + \tau_{\text{err}}^2),$$

where $\tau_{\text{fine}}^2 = \text{finescale.var}$, $\tau_{\text{err}}^2 = \text{error.var}$, and C_0 depends on `sp.type`:

$$C_0(d) = \begin{cases} \sigma^2 \exp(-d/\rho), & \text{exponential,} \\ \sigma^2 \exp(-(d/\rho)^2), & \text{gaussian,} \\ \sigma^2 \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\frac{d}{\rho}\right)^\nu K_\nu\left(\frac{d}{\rho}\right), & \text{matern,} \\ \sigma^2 \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\frac{2\sqrt{\nu}d}{\rho}\right)^\nu K_\nu\left(\frac{2\sqrt{\nu}d}{\rho}\right), & \text{matern2,} \\ \sigma^2 \left(1 - \frac{3}{2}\frac{d}{\rho} + \frac{1}{2}\left(\frac{d}{\rho}\right)^3\right) \mathbf{1}(d < \rho), & \text{spherical.} \end{cases}$$

Here $\nu > 0$ is the Matérn smoothness parameter and $K_\nu(\cdot)$ is the modified Bessel function of the second kind.

For setting 59, the call is

$$\text{simple.cov.sp}(D, \text{"matern"}, (\sigma^2, \rho_i), 0, \nu_i, 0) \quad \text{with } \sigma^2 = 1.$$

Therefore the script uses the unit-variance Matérn correlation kernel

$$R_i(d) = \frac{2^{1-\nu_i}}{\Gamma(\nu_i)} \left(\frac{d}{\rho_i} \right)^{\nu_i} K_{\nu_i} \left(\frac{d}{\rho_i} \right), \quad d > 0,$$

with the diagonal convention $R_i(0) = 1$. The implementation sets the diagonal through the limiting value at $d = 0$, which is why the code replaces NaN values by $\sigma^2 = 1$.

5 From the Matérn Kernel to the Draw-Specific Correlation Matrix

For each posterior draw $i \in \{1, \dots, 5000\}$, the script extracts three draw-specific spatial parameters:

$$\rho_i, \quad \nu_i, \quad \gamma_i.$$

Define the Matérn correlation matrices

$$R_i^{oo} = R_i(D_{22}) \in \mathbb{R}^{n_s \times n_s}, \quad R_i^{pp} = R_i(D_{11}) \in \mathbb{R}^{n_p \times n_p}, \quad R_i^{po} = R_i(D_{12}) \in \mathbb{R}^{n_p \times n_s}.$$

The code then constructs the observed-site correlation matrix as

$$C_i = (1 - \gamma_i)I_{n_s} + \gamma_i R_i^{oo}.$$

This is exactly what the code implements by first computing $\gamma_i R_i^{oo}$ and then forcing the diagonal to be 1.

Likewise, the prediction-side covariance pieces are

$$S_{11,i} = (1 - \gamma_i)I_{n_p} + \gamma_i R_i^{pp}, \quad S_{12,i} = \gamma_i R_i^{po}.$$

The matrix named `prec` in the code is

$$\text{prec}_i = C_i^{-1}.$$

Why `preci` changes with i

The prediction loop runs over $i = 1, \dots, n_{\text{reps}}$. In each draw, ρ_i , ν_i , and γ_i change, so the matrices

$$R_i^{oo}, \quad R_i^{pp}, \quad R_i^{po}, \quad C_i, \quad C_i^{-1}$$

all change as well.

In the current fitted object for setting 59, the posterior draws vary roughly over the following ranges:

$$\rho_i \text{ quantiles (5\%, 50\%, 95\%)} \approx (0.310, 0.388, 0.490),$$

$$\nu_i \text{ quantiles (5\%, 50\%, 95\%)} \approx (0.258, 0.288, 0.331),$$

$$\gamma_i \text{ quantiles (5\%, 50\%, 95\%)} \approx (0.899, 0.924, 0.943).$$

This matters because:

- Larger ρ_i makes spatial dependence decay more slowly with distance, so distant observed sites can influence a prediction more strongly.
- Larger ν_i changes the local smoothness of the Matérn field and therefore changes the shape of the correlation matrix.
- Larger γ_i pushes C_i farther away from the identity matrix, which means the observed residuals are treated as more spatially coherent and the kriging correction becomes stronger.

An important code-level detail is that $\text{prec}_i = C_i^{-1}$ depends only on $(\rho_i, \nu_i, \gamma_i)$, not on t . Therefore, within a fixed posterior draw i , the same inverse correlation matrix is reused for all 31 days. Day-to-day differences then come from the CMAQ covariate values, the partition assignments, the partition precisions, and the imputed observed responses.

6 Partition Membership and Precision Scaling

Setting 59 uses $K = 6$ partitions. For each posterior draw i and day t , the fitted object contains a 6×2 knot matrix

$$\mathcal{K}_{i,t} \in \mathbb{R}^{6 \times 2}.$$

The helper `mem(s, knots)` assigns each location to the nearest knot. Hence the observed-site and prediction-site memberships are

$$g_{i,t}^o(a) \in \{1, \dots, 6\}, \quad a = 1, \dots, n_s,$$

$$g_{i,t}^p(b) \in \{1, \dots, 6\}, \quad b = 1, \dots, n_p.$$

The fitted object also contains partition-specific precisions

$$\tau_{i,k,t} > 0, \quad k = 1, \dots, 6.$$

These are expanded to site-level quantities by lookup:

$$\tau_{i,t}^o(a) = \tau_{i, g_{i,t}^o(a), t}, \quad \tau_{i,t}^p(b) = \tau_{i, g_{i,t}^p(b), t}.$$

Now define the diagonal scaling matrices

$$D_{i,t}^o = \text{diag}(\sqrt{\tau_{i,t}^o(1)}, \dots, \sqrt{\tau_{i,t}^o(n_s)}),$$

$$\Sigma_{i,t}^p = \text{diag}\left(\frac{1}{\sqrt{\tau_{i,t}^p(1)}}, \dots, \frac{1}{\sqrt{\tau_{i,t}^p(n_p)}}\right).$$

These are exactly the code objects `taug.t = sqrt(taug[, t])` and `siggp = 1 / sqrt(tau[gp, t])`.

7 The Conditional Prediction Formula

For posterior draw i and day t , the script loads

$$y_{i,t}^o \in \mathbb{R}^{n_s},$$

where $y_{i,t}^o$ is the imputed observed response vector stored in `fit$y`. This is not the raw observation vector; it is the completed response vector carried forward from the fitting stage.

Because setting 59 is non-skew, the mean term is just

$$\mu_{i,t}^o = X_{o,t}\beta_i, \quad \mu_{i,t}^p = X_{p,t}\beta_i.$$

The observed residual vector is

$$r_{i,t}^o = y_{i,t}^o - \mu_{i,t}^o.$$

The code then standardizes this residual by the partition precisions:

$$u_{i,t}^o = D_{i,t}^o r_{i,t}^o.$$

The central conditional covariance object is

$$Q_i = S_{11,i} - S_{12,i}C_i^{-1}S_{12,i}^\top,$$

which is the Schur complement appearing in Gaussian conditional distributions. In the code this is the object named `corp`.

The predictive mean at unknown locations is

$$m_{i,t}^p = \mu_{i,t}^p + \Sigma_{i,t}^p S_{12,i} C_i^{-1} u_{i,t}^o.$$

This is the line

```
mup <- xp.beta + siggp * s.12.22.inv %*% (taug.t * res[, t]).
```

The predictive covariance is

$$V_{i,t}^p = \Sigma_{i,t}^p Q_i \Sigma_{i,t}^p.$$

Therefore the actual prediction law used by the script is

$$Y_{i,t}^p \mid y_{i,t}^o, \theta_i \sim \mathcal{N}(m_{i,t}^p, V_{i,t}^p),$$

where θ_i denotes all parameters loaded from posterior draw i .

8 How the Script Draws One Realization

Let L_i be a Cholesky factor such that

$$Q_i = L_i L_i^\top.$$

The code computes such a factor from `corp`; if Cholesky fails, it falls back to an eigen-based square root.

If $\varepsilon_{i,t} \sim \mathcal{N}(0, I_{n_p})$, then the script draws

$$Y_{i,t}^p = m_{i,t}^p + \Sigma_{i,t}^p L_i \varepsilon_{i,t}.$$

This is the exact meaning of

```
yp[, t] <- mup + siggp * t.corp.sd.mtx %*% rnorm(np).
```

So the script is not merely computing a kriging mean. It is drawing a full posterior predictive sample for every day and every prediction location.

9 What Changes Across the Posterior Draw Index

Each posterior draw i changes the extrapolation in three coupled ways:

1. The regression term changes through β_i , which alters the CMAQ baseline $\mu_{i,t}^p$.
2. The spatial borrowing pattern changes through $(\rho_i, \nu_i, \gamma_i)$, which changes C_i , $S_{11,i}$, $S_{12,i}$, and therefore $\mathbf{prec}_i = C_i^{-1}$.
3. The scale of the response changes through the day-specific partition precisions $\tau_{i,k,t}$ and the day-specific knot assignments $\mathcal{K}_{i,t}$.

Consequently, the predictive distribution at a fixed unobserved location is not based on a single deterministic set of kriging weights. It is averaged over 5000 distinct spatial systems:

$$\{\beta_i, \rho_i, \nu_i, \gamma_i, \tau_{i,\cdot,\cdot}, \mathcal{K}_{i,\cdot}, y_{i,\cdot}^o\}_{i=1}^{5000}.$$

10 Final Stored Object

For each posterior draw i , each prediction cell b , and each day t , the script stores one predictive sample:

$$\mathbf{y}.\mathbf{pred}[i, b, t] = Y_{i,t}^p(b).$$

Hence the saved object is

$$\mathbf{y}.\mathbf{pred} \in \mathbb{R}^{5000 \times 2632 \times 31}.$$

This means the output is a Monte Carlo representation of the full posterior predictive distribution on the southeast grid, not a single map or a single plug-in predictor.

11 One-Sentence Summary

The prediction code takes each posterior draw from the fitted model, rebuilds the draw-specific spatial correlation matrix C_i , inverts it to obtain $\mathbf{prec}_i$, uses observed-site residuals to form a conditional Gaussian update, rescales that update by partition-specific precisions, and then samples ozone values on previously unobserved CMAQ grid cells.